

Hotel Reservation Analysis Dashboard Report

Project Title

Hotel Reservation Analysis using Power BI

1. Introduction

The Hotel Reservation Analysis Dashboard is developed using Power BI to analyze hotel booking data and provide meaningful business insights. The dashboard helps hotel management understand booking trends, customer behavior, cancellation patterns, and pricing performance.

The analysis allows decision-makers to identify important factors affecting hotel reservations and improve operational efficiency through data-driven decisions.

2. Objectives

The main objectives of this project are:

- Analyze hotel booking trends.
 - Study customer reservation behavior.
 - Identify booking cancellation patterns.
 - Monitor revenue and room pricing.
 - Understand customer segmentation.
 - Support management in strategic decision making.
-

3. Dashboard Overview

The dashboard consists of four interactive pages.

Page 1: Executive Overview

Purpose

This page provides an overall summary of hotel reservations using key performance indicators (KPIs) and booking trends.

KPIs Included

- Total Bookings
- Cancelled Bookings
- Cancellation Rate
- Average Room Price
- Average Lead Time
- Repeat Guest Rate
- Total Guests

Visualizations

- KPI Cards
- Monthly Booking Trend (Line Chart)
- Market Segment Distribution (Donut Chart)
- Interactive Filters (Slicers)

Insights

This page enables users to quickly evaluate the hotel's overall booking performance and identify general trends.

Page 2: Cancellation Analysis

Purpose

This page focuses on understanding why bookings are cancelled.

Visualizations

- Cancellation by Lead Time
- Cancellation by Room Type
- Cancellation by Meal Plan
- Cancellation by Market Segment

Insights

The dashboard helps identify:

- Customer groups with higher cancellation rates.
 - Room categories that experience more cancellations.
 - Meal plans associated with cancellations.
 - Market segments requiring improved retention strategies.
-

Page 3: Revenue and Pricing Analysis

Purpose

This page evaluates pricing trends and hotel revenue performance.

KPIs

- Average Room Price
- Revenue-related Metrics
- Pricing Indicators

Visualizations

- Monthly Average Room Price
- Average Price by Room Type
- Price vs Cancellation Analysis
- KPI Cards

Insights

Management can identify:

- Seasonal pricing trends.

- Revenue generated from different room categories.
 - Relationship between room pricing and booking cancellations.
-

Page 4: Customer Insights

Purpose

This page analyzes customer characteristics and booking behavior.

KPIs

- Customer-related Metrics
- Repeat Guest Statistics
- Guest Distribution

Visualizations

- Customer Segment Analysis
- Repeat vs New Guests
- Guest Composition
- Market Distribution

Insights

This page helps understand:

- Customer demographics.
 - Loyalty of repeat customers.
 - Distribution of different customer segments.
 - Opportunities for improving customer retention.
-

4. Features of the Dashboard

- Interactive slicers
- KPI Cards
- Line Charts
- Column Charts

- Bar Charts
 - Donut Charts
 - Dynamic filtering
 - User-friendly layout
-

5. Business Insights

The dashboard provides several valuable business insights:

- Monitor overall hotel performance through KPI indicators.
 - Identify periods with higher booking demand.
 - Understand customer booking behavior.
 - Detect factors contributing to cancellations.
 - Compare room pricing across different categories.
 - Improve pricing strategies based on booking trends.
 - Increase customer retention by analyzing repeat guests.
 - Support better resource planning and revenue management.
-

6. Benefits of the Dashboard

Using this dashboard enables hotel management to:

- Make informed business decisions.
 - Improve customer satisfaction.
 - Reduce booking cancellations.
 - Optimize room pricing.
 - Increase hotel revenue.
 - Monitor business performance in real time.
-

7. Tools Used

- Microsoft Power BI
- Hotel Reservation Dataset
- Power Query

- DAX Measures
 - Data Visualization Techniques
-

8. Conclusion

The Hotel Reservation Analysis Dashboard effectively transforms raw reservation data into meaningful visual insights. By combining interactive reports with KPIs and analytical charts, the dashboard enables hotel management to monitor bookings, analyze cancellations, understand customer behavior, and optimize pricing strategies. Overall, the project demonstrates how Power BI can be used to support data-driven decision-making in the hospitality industry.